

# Tong Statistical Field Theory — Sheet 2: Correlations and Scaling

*D. Tong, Statistical Field Theory, Example Sheet 2; 3 worked problems.*

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*These prompts paraphrase David Tong’s Statistical Field Theory, Example Sheet 2 (November 2018). Problem numbers, starred status, equations, and guidance follow the official sheet.*

## 1. Sheet 2, Problem 1: Ornstein–Zernicke Correlations

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The Gaussian two-point function is

$$G(x) = \int \frac{d^d k}{(2\pi)^d} \frac{e^{-ik \cdot x}}{\gamma k^2 + \mu^2} = \frac{1}{\gamma} \int \frac{d^d k}{(2\pi)^d} \frac{e^{-ik \cdot x}}{k^2 + \xi^{-2}}, \quad \xi^2 = \frac{\gamma}{\mu^2}. \quad (1.1)$$

Explain why it depends only on  $r = |x|$ . Establish the parameter integral

$$\frac{1}{k^2 + \xi^{-2}} = \int_0^\infty dt e^{-t(k^2 + \xi^{-2})}, \quad (1.2)$$

perform the Gaussian momentum integral, and express the result, up to an overall factor, as

$$G(r) \sim \int_0^\infty dt e^{-S(t)}, \quad S(t) = \frac{r^2}{4t} + \frac{t}{\xi^2} + \frac{d}{2} \log t. \quad (1.3)$$

Use a saddle-point approximation, stating the saddle equation and quadratic prefactor, in the regimes  $r \ll \xi$  and  $r \gg \xi$ . Recover the Ornstein–Zernicke behaviours

$$G(r) \sim r^{-(d-2)} \quad (r \ll \xi), \quad G(r) \sim \frac{e^{-r/\xi}}{r^{(d-1)/2}} \quad (r \gg \xi). \quad (1.4)$$

**Solution.** Since the free energy respects global  $SO(d)$  symmetry, the correlation function must also be invariant under rotation, and hence can only depend on  $r$ . The parameter integral is easy to see, and plugging into the correlation function, we find

$$\begin{aligned} G(r) &\sim \int d^d k \int_0^\infty dt e^{-i\mathbf{k} \cdot \mathbf{x} - t(k^2 + \xi^{-2})} \\ &= \int d^d k \int_0^\infty dt e^{-(\sqrt{t}\mathbf{k} + i\mathbf{x}/2\sqrt{t})^2 - t\xi^{-2} - r^2/4t} \\ &\sim \int_0^\infty dt \frac{e^{-t\xi^{-2} - r^2/4t}}{t^{d/2}}. \end{aligned} \quad (1.5)$$

The saddle point is at

$$S'(t_*) = 0 \implies t_* = \frac{r^2}{d \pm \sqrt{d^2 + 4r^2/\xi^2}}. \quad (1.6)$$

Since  $t > 0$ , we take the plus sign. The correlation function becomes

$$G(r) \sim \int_0^\infty dt e^{-S(t_*) - (t-t_*)^2 S''(t_*)/2} \sim \frac{e^{-S(t_*)}}{\sqrt{S''(t_*)}}. \quad (1.7)$$

For  $r \ll \xi$ , we have  $t_* \sim r^2$ , and hence

$$G(r) \sim r^{-(d-2)}. \quad (1.8)$$

For  $r \gg \xi$ , we have  $t_* \sim r\xi$ , and hence

$$G(r) \sim \frac{e^{-r/\xi}}{r^{(d-1)/2}}. \quad (1.9)$$

## 2. Sheet 2, Problem 5\*: Scaling in a Lifshitz Theory

Separate  $x = (x, \mathbf{y})$  with  $\mathbf{y} \in \mathbb{R}^{d-1}$ , and consider the free energy

$$F[\phi] = \int d^d x \left[ \frac{1}{2} (\partial_x \phi)^2 + \frac{1}{2} (\nabla_y^2 \phi)^2 + \frac{1}{2} \mu_0^2 \phi^2 \right]. \quad (2.1)$$

Write it in Fourier space using momenta  $(k, \mathbf{q})$ . Under  $k' = \zeta k$ ,  $\mathbf{q}' = \zeta^a \mathbf{q}$ , and  $\phi'_{\mathbf{k}'} = \zeta^{-b} \phi_{\mathbf{k}}$ , determine  $a, b$  by normalising both gradient terms, and find the running  $\mu^2(\zeta)$ . In position space, obtain the Gaussian scaling dimension  $\Delta_\phi$ . For an interaction  $\int d^d x g_n \phi^{2n}$ , find the dimension of  $g_n$  and show that  $\phi^4$  is relevant for  $d < 7$  and irrelevant for  $d > 7$ .

**Solution.**

$$F[\phi_{\mathbf{k}}] = \int \frac{d^d k}{(2\pi)^d} \frac{1}{2} (k^2 + \mathbf{q}^4 + \mu_0^2) \phi_{\mathbf{k}} \phi_{-\mathbf{k}} = \int \frac{d^d k'}{(2\pi)^d} \zeta^{-1-a(d-1)} \frac{1}{2} (\zeta^{-2} k'^2 + \zeta^{-4a} \mathbf{q}^4 + \mu_0^2) \zeta^{2b} \phi'_{-\mathbf{k}'} \phi'_{\mathbf{k}'}. \quad (2.2)$$

In order to normalise both gradient terms, we must have

$$a = \frac{1}{2}, \quad b = \frac{d+5}{4}. \quad (2.3)$$

Hence, we find

$$\mu^2(\zeta) = \zeta^2 \mu_0^2. \quad (2.4)$$

In position space, we have

$$\phi'(\mathbf{x}') = \int \frac{d^d k'}{(2\pi)^d} \phi'_{\mathbf{k}'} e^{i\mathbf{k}' \cdot \mathbf{x}'} = \zeta^{1+a(d-1)-b} \int \frac{d^d k}{(2\pi)^d} \phi_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}} = \zeta^{\Delta_\phi} \phi(\mathbf{x}) \implies \Delta_\phi = \frac{d-3}{4}. \quad (2.5)$$

For an interaction, we have

$$\int d^d x g_n \phi^{2n}(\mathbf{x}) = \int d^d x' \zeta^{1+a(d-1)} g_n \zeta^{-2n\Delta_\phi} \phi'^{2n}(\mathbf{x}') = \int d^d x' g_n(\zeta) \phi'^{2n}(\mathbf{x}'), \quad (2.6)$$

which implies

$$g_n(\zeta) = \zeta^{-n(d-3)/2+d/2+1/2} g_n. \quad (2.7)$$

For  $n = 2$ , we have

$$g_2(\zeta) = \zeta^{-(d-7)/2} g_2, \quad (2.8)$$

and hence it is relevant for  $d < 7$ .

## 3. Sheet 2, Problem 7: Emergent Rotational Symmetry

A microscopic cubic lattice in  $d$  dimensions has a local order parameter  $\phi(x)$  with  $\phi \mapsto -\phi$  symmetry. Identify the lowest-order interaction allowed by the lattice rotations but forbidden by full  $SO(d)$  rotations. Use scaling to explain why full rotational invariance can emerge at long distances.

**Solution.** The leading order contributing interaction is

$$\int d^d x \lambda \sum_{i=1}^d \phi \partial_i^4 \phi. \quad (3.1)$$

The coupling constant now has a length dimension  $[\lambda] = 2$ , which is irrelevant for all  $d$ .